

Modelling live social interactions and indirect social influence affecting the adoption of transport technologies and services: a bike sharing case study

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1. *Motivation and objectives*

Social influence is an important factor in the decision-making process of an individual [1, 2]. Especially in contexts of adoption of new technologies and services, it appears reductive to investigate individual's behaviours and responses without considering the inherent social network influence, which can help for a better understanding of the dynamics behind their choices. The processes generated by social influence have been extensively analysed in other disciplines, such as sociology, social psychology and economics. Revising and resuming these studies, Axsen and Kurani [3] conceptualised the dynamics of social interaction processes on intentions and adoptions of new technologies. Another important social influence process discussed in their previous review paper, Axsen and Kurani [4] is the conformity process. Individuals perceive how other people behave and tend to conform under the pressure of subjective and social norms [5]. In fact, norms can influence cognitive processes which generate the intention to a certain behaviour and can also affect the decision making process of the individuals. Besides social interaction and conformity processes, Ng [6] defined an additional level of indirect social influence which is characterised by elusive information and absence of resistance and reinforcement. This influence is very difficult to explain as the individual are not even aware of it (i.e. when a person develops similar attitudes to most of his/her peers).

In transport research, previous methodologies have mainly focused on the inclusion of conformity in quantitative models, i.e. discrete choice models, and have emphasised the analyses of user behaviours at social level, i.e.

25 friends, family and colleagues, with respect to simplified analyses at the in-
26 dividual level. Although there have been significant research advancements,
27 it is yet not clear how to quantify and incorporate in a suitable modelling
28 methodology different social influence effects. For instance, the previous
29 studies lack a modelling methodology that can simultaneously capture real
30 social interaction processes and indirect social influence effect (i.e. the in-
31 fluence described by Ng [6]).

32 Starting from two previous works, Manca et al. [7] and Manca et al. [8]
33 which respectively discussed how to quantify the real/live social interactions
34 and the indirect influence generated by peers' attitudes in a social network,
35 this paper shows an analytical framework to account simultaneously for both
36 processes which can impact the individual's awareness, assessment and self-
37 concept (i.e. the layers of the 'Reflexive Layer of Influence' construct [3]).
38 The proposed analytical framework helps analyse whether the social influ-
39 ence affects the utility of the alternatives, also playing a role in confounding
40 the modified perception of the observable variables and the available alter-
41 natives.

42 *1.1. Direct social influence measures*

43 The direct social influence measures are a consequence of social interac-
44 tions. These measures are conceptually based on the framework developed
45 by Axsen and Kurani [3]. The RLI framework describes three perspectives
46 through the processes (or layers) of social influence which are a direct effect
47 of social interactions within a social network:

- 48 • Diffusion: a process of basic understanding of technology in which
49 awareness of this technology (or service) is diffused during social in-
50 teraction.
- 51 • Translation: a process of technology/service evaluation in which the
52 information is translated into an evaluation of benefits and drawbacks,
53 which can also be translated by each actor differently.
- 54 • Reflexivity: a process of self-concept influence in which discussion
55 about the new technology/service can help one or more of the in-
56 dividuals to reflect upon their own values, lifestyle or self-concept.

57 Besides the direct social interaction processes, the conformity process is
58 included into the general analytical framework because conformity is not
59 completely divisible from social interactions. This helps to give a clearer
60 and more comprehensive overview of the overall social influence processes.

61 1.2. Indirect social influence measures

62 The indirect social influence measures are defined by the interactions
63 between two different types of information: psychometric measures that
64 characterise latent motivational characteristics of the individual, and the tie
65 strengths [8]. The psychometric measures refer to the Theory of Planned Be-
66 haviour (TPB) by Ajzen [9] for the definition of latent characteristics. This
67 theory assumes that the intentions behind a behaviour capture motivational
68 factors. These intentions indicate the degree of people’s willingness to try
69 and the degree of effort that is planned to be practised in order to perform
70 the behaviour [9]. A behavioural intention is formed with the combination
71 of three elements: attitudes, subjective norms towards the behaviour and
72 the perceived difficulty to perform the behaviour.

73 2. Data

74 The case study analysed in this paper refers to the intention within a
75 cohort of students to use a pro-environmental transport modal service (i.e.
76 bike sharing) during a public transport strike [7]. Two different types of sur-
77 vey were performed in 2018 within the student cohort to gather the needed
78 information: a stated preference survey and an attitudinal survey. The
79 targeted SP experiment was designed to explore how alternative attributes
80 affect the individuals’ choice in the hypothetical scenario. Information on
81 normative conformity (i.e. the hypothetical adoption rate within the so-
82 cial network) and real social interactions were also collected. In particular,
83 for the former, a group discussion among the respondents was undertaken
84 between two subsequent stated preference experiments, SP1 and SP2. The
85 group discussion enabled us to investigate cognitive and interpersonal mech-
86 anisms that can generate awareness on the pros and cons of cycling and bike
87 sharing. Moreover, considering the “Reflexive Layers of Influence” (RLI)
88 framework, the processes generated by social interactions (diffusion, trans-
89 lation and reflexivity) are measured and quantified.

90 In the attitudinal survey, the students gave information on Theory of
91 Planned Behaviour constructs (attitudes, subjective norms and perceived
92 behavioural controls concerning cycling). Moreover, respondents were specif-
93 ically asked to name someone in the class they interacted with. They
94 shared information about their social relationships with each person that
95 was named. Specifically, they stated their social proximity to the contact
96 (‘not close’, ‘casual acquaintance’, ‘close’, ‘very close’), frequency of conver-
97 sion with the classmate about commuting between home and university and
98 other occasions of travelling within London.

99 3. Modelling Methodology

100 The proposed modelling framework is complex and built upon the clas-
101 sical discrete choice model formulation, which allows us to investigate the
102 choice behaviour of individuals and how this behaviour can change during
103 the decision-making process. Different techniques are then used to incorpo-
104 rate social influence and latent attitudinal constructs in the discrete choice
105 model framework.

106 The measures of real/live social interactions and social influence pro-
107 cesses (diffusion, translation and reflexivity) are included in a specific choice
108 modelling formulation to evaluate their effect on the individuals' choice.
109 Specifically, the real social interaction is evaluated by incorporating the in-
110 ertia and the propensity to change generated during the group discussion
111 [7].

112 About the quantification of indirect social influence, as in Manca et al.
113 [8], first, an exploratory factor analysis is used to identify these latent con-
114 structs from the psychometric indicators available from the survey data¹.
115 Having identified the subset of indicators specific to each latent construct,
116 a cluster analysis is used on such indicators to group respondents with sim-
117 ilar indicator levels for specific factors. Subsequently, the social influence
118 variable is specified by interacting clusters with social relationship measures
119 (i.e. the social proximity). In order to understand whether individuals with
120 a specific attitude can influence peers in their social network, the individ-
121 ual's peer attitude (IPA) can be included in different components of a Hybrid
122 Choice Model.

123 3.1. The joint hybrid choice model accounting for direct and indirect social 124 influence

125 A Hybrid Choice Model formulation that enables us to account for all
126 the individual attitudes and the individual's social network effects has been
127 used. This model is estimated using a joint SP1/SP2 dataset, incorporating
128 the latent variables and including direct and indirect measures of social
129 influence and social interactions. Its Choice Model Component, whose utility

¹In this specific case study, the identified latent constructs are "Propensity towards cars", "Propensity towards cycling", "Cycling ease" and "Environmental concern". The model shown in this paper only includes the "Latent propensity towards cars" variable which is the only latent variable returning significant results with the model estimations. The statements characterising this construct are: "I would not like to cycle when it rains", "In the city where you previously lived having a car is a must" and "In the city where you previously lived most people drive".

function is associated with alternative a in the stated preference task choice
 $t = [1, \dots, T]$ by the individual i , can be specified as:

$$\begin{aligned} U_{ait}^{SP1} &= V_{ait}^{SP1} + \beta_{aAtt}^{SP1} Att_i + \beta_{SN}^{SP1} SN_{ait}^{SP1} + \xi_{ai}^{SP1} + \varepsilon_{ait}^{SP1} \\ U_{ait}^{SP2} &= \phi(V_{ait}^{SP2} + \beta_{aAtt}^{SP2} Att_i + \beta_{SN}^{SP2} SN_{ait}^{SP2} + \beta_{RLI}^{SP2} RLI_{ait}^{SP2} - I_{ait}^{SP2} + \xi_{ai}^{SP2} + \varepsilon_{ait}^{SP2}) \end{aligned} \quad (1)$$

where $V_{ait} = ASC_a + \beta_{aX} X_{ait} + \beta_{aS} S_i$ is the systematic utility, including alternative characteristics X_{ait} and socioeconomic variables S_i ; SN_{ait} is the effect of the social norm variable capturing the social conformity through information on the adoption rate within the social network; RLI_{ait} is the vector of RLI variables and I_{ait} , the inertia effect, defined in accordance to the general formulation by [10]:

$$I_{ait}^{SP2} = \theta_{ait}^{SP2} (\dot{V}_{kit}^{SP1} - \dot{V}_{ait}^{SP1}) \quad (2)$$

which implies that, during SP2 choice tasks, the individual i also compares the systematic utility regarding the characteristics of the alternative a in SP1, $\dot{V}_{ait}^{SP1}$, to the systematic utility regarding the characteristics of the alternative k actually chosen in SP1 choice tasks, $\dot{V}_{kit}^{SP1}$. In this case, θ_{ait}^{SP2} can vary systematically, randomly and as a function of the group's perception variables of the transport technology specified as $\theta_i = \theta + \delta_i \sigma_\theta - (\theta_{SI}^{St} St_{iG}^+ + \theta_{SI}^W W_{iG}^-)$.

The Structural Model Component associates the latent variable to socioeconomic characteristics S' of the individual i . Moreover, the IPA variable is included to take into account the indirect social influence generated by the peers' attitudes (i.e. the peers' propensity towards cars):

$$Att_i = c + \delta S'_i + \psi IPA_i + \gamma_i \quad (3)$$

where δ and ψ are the parameters associated with the socioeconomic characteristics and the IPA respectively, c is the intercept and γ_i is the error term, assumed to be normally distributed $N(0, \sigma_\gamma)$.

Finally, the specification of the measurement model component, which links the latent variable to the indicators I_{fi} through f equations and for each individual i , is:

$$I_{fi} = d_f + \theta_f Att_i + \nu_{fi}, \quad \text{with } f = 1, \dots, F \quad (4)$$

where θ_f is the coefficient characterising the latent variable, d_f is the intercept and ν_{fi} is the error term, assumed to be normally distributed $N(0, \sigma_\nu)$.

158 4. Results

159 The results of the hybrid choice model including all the different social
160 influence processes are shown in Table 1.

161 4.1. Choice Model Component

162 The choice model component includes all the variables of LoS, the inertia
163 effect, the normative conformity and the latent variable. The parameters
164 regarding the Level-of-Service of the alternatives shared taxi (*Taxi*), bike
165 sharing with docking station (*BSd*) and dockless bike sharing (*BSw*) and
166 the weather conditions are all statistically significant with a confidence level
167 above 95%. Regarding the social norm variable included in the model as
168 generic in *both BS*, the higher the hypothetical adoption rate of peers was,
169 the greater the positive effect of the variable on bike sharing preference.

170 The inertia variable is very significant and positive for *BSw*, indicating
171 the presence of inertia from SP1 to SP2. Hence, if people had previously
172 chosen a mode other than *BSw*, they would again prefer that mode over
173 *BSw*. The coefficients of the interaction between inertia and perceptions are
174 similar in magnitude and significance but opposite sign suggesting that the
175 benefit of ‘avoiding congestion’ reduces the inertia effect and consequently
176 increases the probability of choosing *BSw* while the drawback of ‘safety
177 concern’ increases the inertia effect and, therefore, reduces the propensity
178 to change to another mode during the SP2 experiment. No significant results
179 were found for the inertia variable included in *Taxi* and *BSd*.

180 The different measures of RLI are included as generic in *both BS* of
181 the SP2 part but only the translation variable among all the RLI tested is
182 significant. The negative sign of this variable’s parameter suggests that the
183 respondents do not consider changing mode as high beneficial for themselves.

184 The latent variable (*‘Latent propensity towards cars’*) is included as
185 generic in the utility of *both BS*, its coefficient is always highly significant at
186 more than 95% confidence and with a negative sign; therefore, it is inferred
187 that an individual characterised by ‘propensity towards cars’ is strongly
188 associated with the decrease of utility of *both BS* with respect to the con-
189 ventional *Taxi*, thus, indirectly boosting the demand for *Taxi*.

190 4.2. Structural Model Component

191 The structural model component indicates the socioeconomic character-
192 istics of the individual with a ‘propensity towards cars’. This individual are
193 very likely to be: male (since the dummy variable associated to this charac-
194 teristic has a positive and very significant coefficient), younger than 25 years

195 as suggested by the highly significant negative coefficient on the dummy vari-
196 able for $\text{Age} \geq 25$, not with high cycling experience (this variable coefficient
197 is in fact negative and significant). In addition, this individual is likely to
198 be part of a social network in which the peers are also inclined towards cars.
199 The ‘propensity towards cars’ attitude of the individual is therefore strongly
200 related with the ‘propensity towards cars’ attitude of his/her peers; this in-
201 directly influences the perception of Bike Sharing utility through the effect
202 of the latent variable included in the choice model component.

203 *4.3. Measurement Model Component*

204 Finally, the results of the measurement model component show signifi-
205 cant coefficients of the latent variable (for indicators 2 and 3, always greater
206 than 95%). This confirms that the results of the exploratory factor analy-
207 sis and the presence of correlation between the indicators that identify the
208 latent variable construct.

Table 1: Complete HCM with direct and indirect influence variables

		HCM	
		Value	Robust t-test
Choice model	LoS	<i>ASC (both BS)</i>	11.10 6.88**
		<i>ASC (Taxi)</i>	9.68 4.58**
		<i>Travel Cost (both BS)</i>	-0.75 -6.41**
		<i>Travel Cost (Taxi)</i>	-0.52 -3.17**
		<i>Travel time (both BS)</i>	-0.12 -2.97**
		<i>Travel time (Taxi)</i>	-0.14 -2.65**
		<i>Walking time_gen (both BS)</i>	-0.17 -7.23**
		<i>Weather light rain (Taxi)</i>	3.09 5.81**
		<i>mu (both BS)</i>	5.00 6.57**
		<i>mu (Taxi)</i>	2.32 6.60**
		<i>phi (Scale factor)</i>	1.08 10.39**
	Normative conformity	<i>Social norms (both BS)</i>	0.38 3.85**
	Inertia & interaction with perceptions	<i>I (BSw_SP2)</i>	0.53 2.15*
		<i>I * Avoid congestion (BSw_SP2)</i>	-0.71 -3.03**
		<i>I * Safety concern (BSw_SP2)</i>	1.05 3.46**
	RLI	<i>Operation - diffusion (both BS_SP2)</i>	0.11 0.37
		<i>Own benefit - translation (both BS_SP2)</i>	-2.54 -7.98**
	Latent construct	<i>Latent - Propensity to cars (both BS)</i>	-0.14 -3.50**
Structural model		<i>Age (>25yo)</i>	-0.16 -2.30*
		<i>High cycling experience</i>	-0.08 -2.29*
		<i>Gender - Male</i>	0.14 2.30*
		<i>IPA (Propensity to cars)</i>	0.50 2.28*
		<i>LV Constant</i>	4.65 40.24**
		<i>LV γ</i>	-1.55 -3.55**
Measurement model		<i>Intercept indicator I2</i>	-9.81 -1.55
		<i>Intercept indicator I3</i>	-17.80 -1.86
		<i>Coefficient indicator I2</i>	2.95 2.26*
		<i>Coefficient indicator I3</i>	4.54 2.29*
		<i>Standard deviation indicator I1</i>	-0.45 -1.88
		<i>Standard deviation indicator I2</i>	-0.24 -2.08*
		<i>Standard deviation indicator I3</i>	-3.96 -17.10**
Results		<i>N.param.</i>	31
		<i>N.obs.</i>	816
		<i>N.draws</i>	2000
		<i>Final LL</i>	-725.15
		<i>Rho2</i>	0.393

** $p - value \leq 0.01$ * $p - value \leq 0.05$

209 5. Conclusions

210 The tested modelling structure enables us to disentangle and quantify
211 both direct and indirect social influence effects in the decision-making pro-
212 cess of the individual and, consequently guarantees a better explanation of
213 the heterogeneity. The combined model adds a further degree of complete-
214 ness in explaining the travel choice behaviours of the interviewed respon-
215 dents. Indeed, it makes it possible to relate at the same time:

- 216 • social influence processes of diffusion, translation and reflexivity
- 217 • social live interactions effects such as awareness of benefits and draw-
218 back of a new technology/service
- 219 • conformity processes related to social norms such as the hypothetical
220 adoption rate in the cohort
- 221 • indirect influence processes related to psychometric factors such as
222 attitudes, norms and perceived behavioural control.

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